Tutorial: Landau Notation

Institute of
Algorithms and Theory

Organisatorial

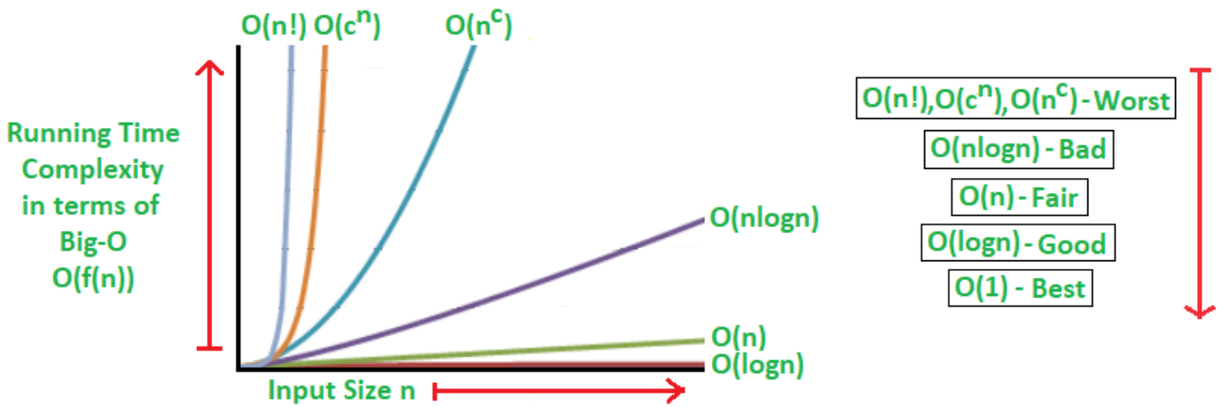
- Switch Tutorials
- TC Tests
- First Theory Sprints
- Questions?

Goal

Landau Notation!

- Fluent with the definitions
- Proving statements like:
  - $\log^2 n = O(n)$
- Get a feeling for asymptotic growth
- Sorting functions: What grows faster?

Landau Notation



one of THE central topics of this class

Landau Notation: Definition

Big-O (O): $O(g) = \{ \mathrm{f} \mid$ there exist constants $c > 0$ and n_0 such that for all $n \geq n_0$,
$$|f(n)| \leq c \cdot |g(n)| \}$$
Interpretation: g is an upper bound on growth rate.

Big-Omega (Ω): $\Omega(g) = \{ \mathrm{f} \mid$ there exist constants $c > 0$ and n_0 such that for all $n \geq n_0$,
$$|f(n)| \geq c \cdot |g(n)| \}$$
Interpretation: g is a lower bound on growth rate.

Big-Theta (Θ): $\Theta(g) = \{ \mathrm{f} \mid$ there exist constants $c_1, c_2 > 0$ and n_0 such that for all $n \geq n_0$,
$$c_1 \cdot |g(n)| \leq |f(n)| \leq c_2 \cdot |g(n)| \}$$
Interpretation: g has asymptotically the same growth rate.

Formally, $O(g), \Omega(g), \Theta(g)$ are **sets** of functions, but one usually misuses notation and writes
$$f = O(g),\ f = \Omega(g), \text{ and } f = \Theta(g)$$

One-sided "equality". $f = O(g)$ does not automatically mean $g = O(f)$

Landau-Notation: Equivalent Characterization

Big-O(O): $f(n) = O(g(n))$ iff

$$\limsup_{n\to\infty} \frac{|f(n)|}{|g(n)|} < \infty$$

Big-Omega (Ω): $f(n) = \Omega(g(n))$ iff

$$\liminf_{n\to\infty} \frac{|f(n)|}{|g(n)|} > 0$$

Theta (Θ): $f(n) = \Theta(g(n))$ iff

$$0 < \liminf_{n\to\infty} \frac{|f(n)|}{|g(n)|} \leq \limsup_{n\to\infty} \frac{|f(n)|}{|g(n)|} < \infty$$

Little-o (o): $f(n) = o(g(n))$ iff

$$\lim_{n\to\infty} \frac{|f(n)|}{|g(n)|} = 0$$

Little-omega (ω): $f(n) = \omega(g(n))$ iff

$$\lim_{n\to\infty} \frac{|f(n)|}{|g(n)|} = \infty$$

Examples: Landau Notation

Show that $3n^2 + 17n + 1000 = \Theta(n^2)$ for both different definitions of the Θ-notation.

Show that $4n \neq \Omega(n^2)$ using the set theoretic definition of the Ω-notation.

Examples: Landau Notation

$$4n \neq \Omega(n^2)$$

What does that mean?

For **all** $n_0, c > 0$ there exists a $n \geq n_0$ such that $4n < c \cdot n^2$.

How can we prove this? Check all combinations of constants c and n_0?

n_0 and c are given (by an adversary). Proof by example is **NOT** enough.

Examples: Landau Notation

Show that $3n^2 + 17n + 1000 = \Theta(n^2)$ for both different definitions of the Θ-notation.

Show that $4n \neq \Omega(n^2)$ using the set theoretic definition of the Ω-notation.

Show that $n^k = O(n^l)$ for two constants $k \leq l$.

Examples: Landau Notation

Prove or disprove the following statements:

- $n^{13} = O(n^{10})$
- $\log n = O(n^2)$
- $4000 \log n = O(n)$
- $\log n = o(n)$
- $n \log n = \Omega(n^2)$
- $20000\, n \log n = \Omega(n^2)$

Examples: Landau Notation

What is the asymptotic growth of the following functions?

Function	Dominant term	O(…)
$7n^3 + 1.5n^2 - 4n + 1000$	$7n^3$	$O(n^3)$
$500n + 100n^{1.5} + 50n \log_{10} n$	$100n^{1.5}$	$O(n^{1.5})$
$1000n \log_2 n + 0.1n(\log_2 n)^2$	$0.1n(\log_2 n\ )^2$	$O(n(\log n)^2$
$50n^{1.5} + 3n^{1.75} + 1000$	$3n^{1.75}$	$O(n^{1.75})$
$2n^3 + 9\sqrt{n} + 2n - 532$	$2n^3$	$O(n^3)$
$0.33 \log_4 n + \log_2 \log_2 n$	$0.33 \log_4 n$	$O(\log n)$
$3n + 7000n^{20} + n^{1001} + 4020n^7$	n^{1001}	$O(n^{1001})$
$n^{100} + 2^n + 232n^{66}$	2^n	$O(2^n)$

Landau Notation: Some Runtime Classes

- $O(1)$ Constant time
- $O(\log n)$ Logarithmic time (Base does **not** matter)
- $O(\log^c n)$ Polylogarithmic time
- $O(n)$ Linear time
- $O(n \log n)$ (Linearithmic time)
- $O(n^2)$ Quadratic time
- $O(n^p)$ Polynomial time
- $O(c^n)$ Exponential time (Base **does** matter)
- $O(n!)$ (Factorial time)

Pseudocode: Runtime

```
algorithmA(n)
  for i = 1; i ≤ n; i++ do
      Print(Hello);
  end for
  return
```

```
algorithmB(n)
  for i = 1; i ≤ n; i++ do
      for j = 1; j ≤ n; j++ do
          Print(Hello);
      end for
  end for
  return
```